

Jihyun Lee

Assistant Professor at the University of North Texas

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Education

Ph.D. in Quantitative Methods

2022

Department of Educational Psychology, University of Texas at Austin

- Dissertation: Handling Missing Covariates in Meta-Regression Using Multilevel Multiple Imputation with Factored Regression Specification

M.A. in Psychometrics and Statistics in Psychology

2016

Department of Psychology, Ewha Womans University

- Thesis: Item Parceling in Structural Equation Modeling - The Integration of Controversies and Proposals

B.S. in Psychology

2014

Department of Psychology, Ewha Womans University

- Magna Cum Laude

Professional Position

Assistant Professor

2022 - Current

Research methodology, Measurement, and Statistics | Department of Educational Psychology | University of North Texas

Research Interests

Meta-Analysis: Handling missing data in meta-analysis, missing covariates in meta-regression

Missing Data Analysis: Multiple imputation (MI), multilevel MI, factored-regression specification approach (substantive model-based) to MI

Latent Variable Modeling: Structural equation modeling (SEM), item parceling, multigroup SEM

Methodological Publications

Lee, J., & Beretvas, N. S. (Accepted). Handling within-study dependence in correlation coefficients and moderators in meta-analytic structural equation modeling, *Research Synthesis Methods*

Lee, J., Beretvas, S. N., & Keller, B. T. (2026). Exploring the Use of Multiple Imputation for Handling Missing Covariates in Meta-Regression with Dependent Effect Sizes. *Multivariate Behavioral Research*, 1–25. <https://doi.org/10.1080/00273171.2026.2673547>

Whittaker, T. A., **Lee, J.**, Dedrick, D., & Muñoz, C. (2024). Meta-analysis of Monte Carlo simulations examining class enumeration accuracy with mixture models. *Psychological Methods*. Advance online publication, 1–14. <https://doi.org/10.1037/met0000716>

Lee, J. & Beretvas, S.N. (2023), Missingness mechanism messiness in meta-analysis: A response to "The importance of plausible data-generating mechanisms in simulation studies". *Research Synthesis Methods*. 14(1) 140–142. <https://doi.org/10.1002/jrsm.1603>

Lee, J. & Beretvas, S. N. (2023) Comparing Methods for Handling Missing Covariates in Meta-Regression. *Research Synthesis Methods*. 14(1) 117-136. <https://doi.org/10.1002/jrsm.1585>

Schauer, J., **Lee, J.**, Diaz, K., & Pigott, T.D. (2022). On the Bias of Complete- and Shifting-Case Meta-Regressions with Missing Covariates. *Research Synthesis Methods*, 13(4) 489-507, <https://onlinelibrary.wiley.com/doi/10.1002/jrsm.1558>

Schauer, J. M., Diaz Yanez, K., Pigott, T. D., & **Lee, J.** (2022). Exploratory Analyses for Missing Data in Meta-Analyses and Meta-Regression: A Tutorial. *Alcohol and Alcoholism*, 57(1) 35-46, <https://doi.org/10.1093/alcalc/agaa144>

Lee, J. & Whittaker, T. A. (2021). The impact of item parcelling on structural parameter invariance in multi-group structural equation modeling. *Structural Equation Modeling: A Multidisciplinary Journal*, 28(5), 684-698, <https://doi.org/10.1080/10705511.2021.1890604>

Lee, J. & Kim, S. Y. (2016). Item Parceling: Understanding and Applying the Principles. *Korean Journal of Psychology: General*. 35(2), 327-353, <https://doi.org/10.22257/kjp.2016.06.35.2.327>

Applied Research Publications

Atit, Peterson, Gilligan-Lee, Hawes, & **Lee**¹ (Accepted 5/8/26). How Spatial Skills Relate to Science Achievement: A Meta-Analytic Review. *Psychological Bulletin*

Patall, E. A., Yates, N., **Lee, J.**, Chen, M., Hamilton, B., Lee, K., Beretvas, S. N., Lin, S., Yang, S. M., Jacobson, N. G., Harris, E., & Hanson, D. J. (2024). A meta-analysis of teachers' provision of structure in the classroom and students' academic competence beliefs, engagement, and achievement. *Educational Psychologist*, 59(1), 42-70. <https://doi.org/10.1080/00461520.2023.2274104>

- Awarded as 2025 *Educational Psychologist* Article of the Year. (APA Division 15's Publication Committee)

Fong, C., **Lee, J.**, Hoff, M. L., Krou, M. R., Johnston-Ashton, K., Gonzales, C., & Beretvas, S. N. (2023). Meta-Analyzing the Factor Structure of the Learning and Study Strategies Inventory. *The Journal of Experimental Education*, 91(2) 380-400. <https://doi.org/10.1080/00220973.2021.2021842>

Atit, K., Power, J. R., Pigott, T., **Lee, J.**, Geer, E. A., Uttal, D. H., Ganley, C. M., & Sorby, S. A., (2022). Examining the Relations Between Spatial Skills and Mathematics Performance: A Meta-Analysis. *Psychonomic Bulletin & Review*, 29, 699-720, <https://doi.org/10.3758/s13423-021-02012-w>, Online supplementary materials available: <https://osf.io/h3w7j/>

Roberts, G. J., Hall, C., Cho, E., Cote, B., **Lee, J.**, Qi B., & Van Ooyik, J. (2022). The state of current reading intervention research for English learners in grades K-2: A best-evidence synthesis. *Educational Psychology Review*. 34, 335-361. <https://doi.org/10.1007/s10648-021-09629-2>

Invited Publications

Lee, J. & Beretvas, S. N. (2020). Comparing Methods for Handling Missing Moderators in Meta-Analysis: A Robustness Study,² *Multivariate Behavioral Research*, 55(1), 149, <https://doi.org/10.1080/00273171.2019.1697862>

Papers in Progress

¹Gilligan-Lee, Hawes, and **Lee** share the third authorship. **Dr. Lee** carried out all analyses and writing methods and results of the study.

²This is an abstract publication as a part of the SMEP annual meeting presentation.

Madson, S. K.*, **Lee, J.**, Martin, S. B., & Jung, M. (Under Review). Understanding Physical Activity During Pregnancy and Postpartum Through Self-Determination Theory: A Systematic Review. *Women & Health*.

Lee, J. (under review) Using multiple imputation for handling missing covariates in meta-regression with interactions: an evaluation and illustration, *Research synthesis methods*

Lee, J. (Co-First author) & Fong, C. (Co-First author) (in progress). Charting pathways between motivational, cognitive, and meta-cognitive strategies: A meta-analytic structural equation modeling approach

Lee, J. (Co-First author), Hosek, B. (Co-First author), Zhang, Y.* (in progress)

Lee, J., (Zhang, Y.*, Flora*, Abena*) (in progress) Recap the practice of meta-regression and missing data handling

Lee, J. (in progress) Missing data and interaction effect in meta-regression

Grants and Contracts

Funded Grants and Contracts

PI, *Funded contract with American Psychological Association (APA) through UNT Grants and Research Administration Management Suite (GRAMS) - GFP0001262, APA* 12/2023 - current
– The project job includes working on research projects with the Center for Psychology in Schools and Education in the Education Directorate in achieving its mission to generate empirical research for how psychological science informs K12 contexts, including performing a series of meta-analysis for the Center.

University of North Texas Faculty First Flight Funds
\$500 2023 Summer

University of North Texas; Travel Award
\$1000 2026

External Grants and Contracts Not Funded or Under Review

Co-Investigator, Title: Developing a non-invasive acoustic tool to regulate blood flow and enhance endothelial cells' release of anticoagulant and antithrombotic factors 2024 Summer submitted

– Agency: W.M. Keck Foundation Not funded
Budget \$1,200,000

Co-PI, Title: Beyond Numbers: Measuring Data Literacy Competency for Grades 9=12 Through Scenario-based Assessment 2023 Fall submitted

– Agency: National Institutes of Health (NIH) Not funded
Budget \$1,379,635

Professional Presentations

*Presenting authors in boldface. Student presenters are indicated with *.*

Lee, J. (2026, June) Modern Modeling Methods, New York

Lee, J. (2026, March 28) TUESAP

Madson, S. K.*, Jung, M., Lee, J., & Martin, S. B. (accepted; 2025, October 15-18) *The role of physical activity in motivation and basic psychological needs in pregnancy and postpartum: A systematic review within the framework of self-determination theory*. Association for Applied Sport Psychology, Montréal, Canada

Lee, J. (2025, July 9-11), *Accounting for Dependency in Your Analysis? Don't Forget It for Missing Data Too*. Society of Research Synthesis Methodology (SRSM), Madison, Wisconsin

[Poster] **Madson, S. K.***, Jung, M., Lee, J., & Martin, S. B. (2025, April 14th) *The role of physical activity in motivation and basic psychological needs in pregnancy and postpartum: A systematic review within the framework of self-determination theory*. Research Showcase at College of Education, University of North Texas

Lee, J. (2024, August) *Broaden Your Research Methodology with Meta-Analytic Structural Equation Modeling*. American Psychological Association (APA), Seattle, WA

Lee, J., & Beretvas, S. N. (2024, June) *Handling within-study dependency in effect sizes and moderator values in meta-analytic structural equation modeling*. SRSM, Amsterdam, Netherlands

Lee, J. (2023, August) *Understand and handle missing covariates in meta-regression*. APA, Washington, DC

Lee, J., Beretvas, S. N., & Keller, B. T. (2023, July) *Handling Missing Covariates in Meta-Regression using Multilevel Multiple Imputation with Factored Regression Specification*, SRSM, Paris, France

Lee, J. (2022, May, invited) *Missing Covariates in Meta-Regression: Considerations and Implications*. Invited to talk at Symposium: "Early Career Spotlight: Increasing Diversity in the Quant Pipeline" at the Association for Psychological Science (APS), Chicago, IL.

Schauer, J., **Lee, J.**, Diaz-Yanez, K., Pigott, T. (2022, April) *On the Bias of Complete- and Shifting-Case Meta-Regressions with Missing Covariates*. Paper presented at the American Educational Research Association (AERA) annual meeting, Systematic Review and Meta-Analysis SIG

Whittaker, T., Lee, J., Dedrick, D., Munoz, C. (2022, April) *Meta-Analysis of Monte Carlo Simulations Examining Class Enumeration Accuracy with Mixture Models*, The Utility of Mixture Modeling to Understanding Heterogeneous Effects in Educational Studies, Paper presented at the AERA annual meeting

Pigott, T., Diaz-Yanez, K., Lee, J., & Schauer, J. (2021, October 22) Invited talk: *Missing Covariates in Meta-Regression: Considerations and Implications*, Online Seminar at Systematic Reviews and Meta-Analysis Methodology, <https://www.srmasig.org/seminar/>

Fong, C., Lee, J., Hoff, M. L., Krou, M. R., Johnston-Ashton, K., Gonzales, C., & Beretvas, S. N. (2021, April). *Extending the Investigation of LASSI's Factor Structure: A Meta-Analytic Structural Equation Modeling Approach*. Paper presented at the AERA annual meeting, Systematic Review and Meta-Analysis SIG (online)

Fong, C., Lee, J., Hoff, M. L., Krou, M. R., Johnston-Ashton, K., Gonzales, C., & Beretvas, S. N. (2020, April). *Extending the Investigation of LASSI's Factor Structure: A Meta-Analytic Structural Equation Modeling Approach*. Paper presented at the American Educational Research Association (AERA) annual meeting, Systematic Review and Meta-Analysis SIG, (Conference Canceled due to pandemic; Online repository available)

[Poster] **Lee, J.** & Beretvas, S. N. (2019 October 9 - 12). *Comparing Methods for Handling Missing Moderators in Meta-Analysis: A Robustness Study*. Poster presented at the Society of Multivariate Experimental Psychology (SMEP) 2019 Annual Meeting, Baltimore, MD.

[Poster] **Lee, J.**, & Whittaker, T. A. (2018, August 9 - 12). *The Impact of Item Parceling on Invariance Tests at the*

Structural Level in Multi-group Structural Equation Modeling. Poster presented at the American Psychological Association (APA) 2018 Annual Convention, Division 5, San Francisco, CA.

Lee, J., & Kim, S.Y. (2016, May). *Using Item Parceling: The Integration and Proposals*. Paper presented at the Korean Psychological Measurement and Evaluation Association, Seoul, South Korea.

Previous Research Experience

Data Analyst/Research Assistant, *School-Based Marginalization, Social-Behavioral Competencies, and Achievement: A Meta-analysis*, Dr. Varner, F. & Dr. Beretvas, S. N., The University of Texas at Austin 8/2020 - 5/2022

- Conduct statistical analyses using R (data cleaning, calculating inter-rater correlation, meta-analytic structural equation modeling).

Data Analyst/Research Assistant, The Meadows Center: For Preventing Educational Risk 6/2021 - 12/2021

- Conduct statistical analyses using R and Mplus (data cleaning, missing data analysis, latent profile analysis, factor mixture modeling) and write a manuscript.

Data Analyst/Research Assistant, *A Meta-Analysis of Classroom Structure Effects on K-16 Students' Academic Motivation, Engagement, and Achievement*, Dr. Patall, E. A. & Dr. Beretvas, S. N., The University of Texas at Austin 8/2019 - 8/2020

- Conducted statistical analyses using R (data cleaning, meta-analysis) and wrote a report and a manuscript
- The Institute of Education Sciences, U.S. Department of Education, through Grant R305A160406 to University of Southern California.

Data Analyst/Research Assistant, *NIH Grant Behavioral Regulation Project*, Dr. Schmitt, M., The University of Texas at Austin 6/2019 - 8/2019

- Conducted statistical analyses using R and SPSS (experimental data analysis, factor analysis, difference-in-difference analysis) and wrote a report

Research Assistant/Statistic Consultant, *Cain Center for Nursing Research*, School of Nursing, The University of Texas at Austin 1/2017 - 8/2017

- Statistical analysis consulting for graduate students

Research Assistant, *Development of Career Aptitude exploration Test (CARAT) for middle and high school students*, Dr. Kim, A.Y., & Dr. Kim, S.Y., Korea Institute of Psychological Aptitude Test (KIPAT) & Ewha Womans University 1/2015 - 12/2015

- Conducted literature review for the test constructs, developed items and designing a test, statistical analyses (including differential item functioning (DIF), confirmatory factor analysis (CFA)), and wrote a technical manual and a report

- Research Assistant**, *Development of Academic Aptitude Test (AAT) for elementary students*, Dr. Kim, A.Y., & Dr. Kim, S.Y., Korea Institute of Psychological Aptitude Test (KIPAT) & Ewha Womans University 12/2013 - 12/2014
- Developed a test, conducted the test to elementary students, conducted statistical analyses (including DIF & CFA), and wrote a technical manual and a report
- Statistical Consultant/Data Analyst**, *Evaluation of Spelling and Grammar in Korean Language*, Dr. Lee, K.G. & Dr. Kim, S.Y., The National Institute of Korean Language 1/2014 - 12/2014
- Provided statistical consulting to the applied researchers, conducted statistical analyses and wrote a report

Instructional Activities

Graduate Courses (Instructor)

EPSY6800. Special Topics: Meta-analysis	S2025
Department of Educational Psychology, University of North Texas (UNT)	
EPSY6270. Structural Equation Modeling	S2026
Department of Educational Psychology, University of North Texas (UNT)	
EPSY6240. Technology in Research (Data Science using R)	F2023, F2025
Department of Educational Psychology, UNT	
EPSY6020. Research Methods in Education	S2024, S2025
Department of Educational Psychology, UNT	
EPSY6010. Statistics for Educational Research	F2022, F2023, F2024, F2025, S2026
Department of Educational Psychology, UNT	
EPSY5210. Educational Statistics	S2023 (online), S2024 (online), Summer 2024 (online), F2024, Summer 2025 (online)
Department of Educational Psychology, UNT	

Graduate Courses (Teaching Assistant)

Missing Data Analysis	S2021
Department of Educational Psychology, University of Texas (UT) at Austin	
Meta-analysis	F2020
Department of Educational Psychology, UT-Austin	
Structural Equation Modeling	S2020; S2019; F2018; S2017
Department of Educational Psychology, UT-Austin	
Multivariate Methods	S2022
Department of Educational Psychology, UT-Austin	
Statistical Analysis for Experimental Data	F2019; F2018; Summer2018
Department of Educational Psychology, UT-Austin	
Correlation and Regression	Summer2020; 2017
Department of Educational Psychology, UT-Austin	
Fundamental Statistics	Summer2020; 2017
Department of Educational Psychology, UT-Austin	
Psychometrics theory and methods	S2019; S2018
Department of Educational Psychology, UT-Austin	

Undergraduate Courses (Assistant)

Data Analysis for the Health Sciences	F2017
Department of Statistics and Data Science, UT-Austin	
Statistics in Market Analysis	F2016
Department of Statistics and Data Science, UT-Austin	

Workshops (Teaching Assistant)

Quantitative Meta-Analysis Workshop, CenterStat	6/15 - 6/18, 2021
Instructor: Beretvas, S. N.	Online
Notre Dame Pier Methods Meta-Analysis Workshop, University of Notre Dame	5/25, 2021
Instructor: Beretvas, S. N.	Online
Meta-Analysis Training Institute (MATI)	7/28 - 8/3, 2019
	Chicaco, IL

Graduate Students Advising

Dissertation Committees Serving On

- Xiaolu Fan (Research, Measurement, and Statistics (RMS) program): 2025 - in progress
- Ashley Shen (Gifted Talent program): 2025 - in progress
- Nino Kokiashvili (RMS): 2024 - in progress

Dissertation Committees Served

- Genea Stewart: 2022 - Summer 2023 defended

Doctoral Qualifying Process Committees Serving On

- Xiaolu Fan: 2024 – Completed (P)

Supervising

- Yan Zhang, a doctoral student in RMS, EPSY, UNT: research assistant (Spring 2025 - Current)
- Xiaolu Fan, a doctoral student in RMS, EPSY, UNT: research assistant (Fall 2024)

Master’s Student Advising

- Sydney Winn: Summer 2025 - Current
- Clara Pereira: Fall 2025 - (Transferred)

Doctoral Student Advising

- Wei Sun: Fall 2025 - Current

Honors & Awards

Scholarships & Fellowships

Graduate School Summer 2022 Fellowship, The University of Texas at Austin	2022
William L. Hays Endowed Fellowship, The University of Texas at Austin	2020-2022
Earl Jennings Educational Psychology Excellence Award, The University of Texas at Austin	2018-2020
Texas New Scholar Fellowship, The University of Texas at Austin	2017-2018
Graduate School Recruitment Fellowship, The University of Texas at Austin	2016-2017
Merit-Based Scholarship for Academic Excellence, Ewha Womans University, Seoul, S. Korea	2014
With a full-tuition waiver	
Dean’s Listed and Merit-Based Scholarship for Academic Excellence, Ewha Womans University, Seoul, S. Korea	2010-2012

Travel Awards

Conference Support Award, UNT \$1,000 to attend Modern Method Modeling conference	2026
Professional Development Award, Dean of the Graduate School and Senior Vice Provost of Academic Affairs, UT Austin	2021, 2022
Presenter Travel Award, Society of Multivariate Experimental Psychology	2019 <i>Baltimore, MD</i>
Professional Development Award, Educational Psychology Department, UT Austin	2018 <i>San Francisco, CA</i>
Ungerleider/Zimbardo Travel Scholarship, American Psychological Foundation	2018 <i>San Francisco, CA</i>

Professional Journal Activities

Research Synthesis Methods Associate Editor; https://www.cambridge.org/core/journals/research-synthesis-methods/information/about-this-journal/editorial-board	2025 - Current
Review of Educational Research Editorial board member; https://journals.sagepub.com/editorial-board/rer	2025 - Current
Campbell Collaboration Editorial board team member: Methods group advisory board; https://www.campbellcollaboration.org/methods/team/	2024 - Current

Manuscript Review

Biometrical Journal Reviewer (1)	2026
Psychological Methods Reviewer (1)	2026
Research Synthesis Methods Reviewer (2 manuscripts)	2023-current
Review of Educational Research Reviewer (2 manuscripts)	2025-current
CNS Drugs Reviewer (1)	2025
Psychology of Aesthetics, Creativity, and the Arts Reviewer (2)	2024-2026
Structural Equation Modeling: A Multidisciplinary Journal Reviewer (1)	2021

Conference paper review

American Educational Research Association (AERA) Annual Meeting, Systematic Reviews and Meta-Analysis SIG	2020, 2022, 2025
Reviewer	
Society of Multivariate Experimental Psychology (SMEP) Annual Meeting	2020
Reviewer: three graduate student poster session submissions	

Professional Membership

Treasurer, Systematic Review and Meta-Analysis SIG, AERA	2025 - present
- 2026 AERA annual meeting. Business meeting presentation	
Member, AERA	present
+ Chaired a session served in AERA annual meeting 2025	
Member, Society of Research synthesis Methodology	2023 - present
Member, American Psychological Association	2023 - 2024